



COURSE HAND OUT

School:	School of Computing				Dept.:	CSE
Course Code	: 22CS102010					
Course Title	: Software Testing and Quality Assurance					
Course Credit Structure	L	T	P	S	C	
	3	-	2	-	4	
Semester	B. Tech VII Semester					
Contact Hours	: 45 Hours					
Instructor	: P.Suresh Kumar Yadav					
Instructor's Email	: Sureshkumar.p@mbu.asia					
Office Hours	: All working days					
Academic Year	: 2026-27					
Date of Issue	: 13-07-2026					

PROGRAM OUTCOMES

On successful completion of the Program, the graduates of B. Tech. (CSE) Program will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

On successful completion of the Program, the graduates of B.Tech.(CSE) program will be able to:

- PSO1:** Use mathematical methodologies to model real-world problems, Employ modern tools and platforms for efficient design and development of computer-based systems.
- PSO2:** Apply adaptive algorithms and methodologies to develop intelligent systems for solving problems from inter-disciplinary domains.
- PSO3:** Apply suitable models, tools and techniques to perform data analytics for effective decision making.
- PSO4:** Design and deploy networked systems using standards and principles, evaluate security measures for complex networks, apply procedures and tools to solve networking issues.

Course Code	Course Title
22CS102010	Software Testing and Quality Assurance
Pre-Requisite	Software Engineering

COURSE DESCRIPTION: This course presents a practical approach to software testing as a sub-discipline of software engineering. It introduces software quality concepts, standards, measurements, and practices that support the production of quality software. It offers a solid foundation in testing fundamentals including test case design, test management, and test measurement strategies, which improve the effectiveness of software test processes.

COURSE OUTCOMES: After successful completion of the course, students will be able to:

- | | |
|------------|--|
| CO1 | Review the similarities and differences in software quality practices between plan-driven and agile teams. |
| CO2 | Integrate software quality assurance practices using Testing Maturity Model (TMM) levels for software development processes. |
| CO3 | Develop a comprehensive software quality and test plan. |
| CO4 | Analyze test cases to support multiple testing goals. |

CO-PO-PSO Mapping Table:

Course Outcome	Program Outcomes												Program Specific Outcomes			
	PO1	PO2	PO3	PO4	PO5	PO 6	PO7	PO 8	PO 9	PO1 0	PO1 1	PO 12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	-	-	-	-	-	-	-	-	-	3	-	3	-
CO2	3	3	-	-	-	-	2	-	-	-	-	-	3	-	3	-
CO3	3	3	3	-	-	-	-	-	-	-	2	-	3	-	3	-
CO4	2	3	3	-	-	-	-	2	-	-	-	-	3	-	-	-
Course Correlat ion Mapping	3	3	3	-	-	-	2	2	-	-	2	-	3	-	3	-

Correlation Level:

3-High

2-Medium

1-Low

COURSE CONTENT

Module 1 – Introduction

(9 periods)

Agile Testing – Traditional vs agile testing - Principles for agile testers – Mind set- Applying principles and values –Adding value.

Module 2 – Team Logistics

(9 periods)

Team Structure - Independent QA Teams - Integration of Testers into an Agile Project - Agile Project Teams - Physical Logistics - Resources - Tester-Developer Ratio - Hiring an Agile Tester - Building a Team - Self-Organizing Team - Involving Other Teams - Every Team Member Has Equal Value - Performance and Rewards.

Module 3 – Transitioning Typical Processes

(9 periods)

Lightweight Processes - Metrics Defect Tracking – Tracking tools - Test Planning - Test Strategy vs. Test Planning - Traceability - Existing Processes and Models - Audits - Frameworks, Models, and Standards.

Module 4 – Test Tool Strategy

(9 periods)

Tools to Elicit Examples and Requirements - Checklists - Mind Maps - Spreadsheets - Mock-Ups - Flow Diagrams - Software-Based Tools - Tools for Automating Tests Based on Examples - Tools to Test below the GUI and API Level - Tools for Testing through the GUI - Strategies for Writing Tests - Testability - Code Design and Test Design.

Module 5 – Tests that Critique the Product

(9 periods)

Introduction to Quadrant 3 - Demonstrations - Scenario Testing - Exploratory Testing - Usability Testing - Behind the GUI - Testing Documents and Documentation - User Documentation - Reports - Tools to Assist with Exploratory Testing.

Total Periods 45

List of Experiments:

Q. No.	Question	POs
1.	Find out defects on the page shown below 	PO1, PO2, PO3

Q. No.	Question	POs
2.	Write test ideas for this Scenario: You are at the grocery store's checkout counter. You have bought five items (x, y, z, a, and b). You make payment and move to the EXIT door.	PO1, PO2, PO7
3.	Write a detailed defect report for this sample defect: After logging into Gmail, it navigates to Google.com	PO1, PO2, PO3, PO11
4.	How user experience can be improved for the following sign-in page? 	PO1, PO2, PO3, PO8

Text Book

1. Lisa Crispin, Agile Testing, Addison Wesley, 2012.

Reference Book:

1. Burnstein, I. (2003). Practical Software Testing. Springer-Verlag. ISBN: 0-387-95131-8.

SOFTWARE/TOOLS:

1. selenium

VIDEO LECTURES:

1. <https://nptel.ac.in/noc/courses/noc19/SEM2/noc19-cs70/>
2. <https://archive.nptel.ac.in/courses/117/106/117106112/>
3. <https://www.softwaretestinghelp.org/>

WEB RESOURCES:

1. <https://www.geeksforgeeks.org/software-testing-manual-testing/>
2. <https://www.javatpoint.com/manual-testing>

PEDAGOGY:

The following pedagogy methods will be used to deliver the course.

- A) Black Board
- B) PPT
- C) Open Discussion
- D) Blended Learning

COURSE DELIVERY SCHEDULE:

S. No	Topic	Contact Hours	CO Mapping	Pedagogy	Resources
MODULE-1: INTRODUCTION (9 Hours)					
1.	Agile Testing	02	C01	PPT, Open Discussion Session	TB1
2.	Traditional vs agile testing	02	C01	Blended Learning	TB1
3.	Principles for agile testers	02	C01	Comparative study	TB1
4.	Mind set	01	C01	PPT and Open Discussion Session	TB1
5.	Applying principles and values	01	C01	PPT	TB1
6.	Adding value.	01	C01	PPT and Open Discussion Session	TB1
MODULE-2: TEAM LOGISTICS(9 Hours)					
7.	Team Structure	01	CO2	PPT	TB1
8.	Independent QA Teams	01	CO2	PPT	TB1
9.	Integration of Testers into an Agile Project	01	CO2	PPT	TB1
10.	Agile Project Teams - Physical Logistics - Resources	01	CO2	PPT	TB1
11.	Tester-Developer Ratio - Hiring an Agile Tester	01	CO2	PPT	TB1
12.	Building a Team - Self-Organizing Team	01	CO2	PPT	TB1
13.	Involving Other Teams	01	CO2	PPT	TB1
14.	Every Team Member Has Equal Value	01	CO2	PPT	TB1
15.	Performance and Rewards.	01	CO2	PPT	TB1
MODULE-3: TRANSITIONING TYPICAL PROCESSES (9 Hours)					
16.	Lightweight Processes	01	CO3	Blended Learning	TB1
17.	Metrics Defect Tracking	01	CO3	PPT	TB1
18.	Tracking tools	01	CO3	PPT	TB1
19.	Test Planning	01	CO3	PPT	TB1
20.	Test Strategy vs. Test Planning	01	CO3	PPT	TB1
21.	Traceability	01	CO3	PPT	TB1
22.	Existing Processes and Models - Audits	01	CO3	PPT	TB1
23.	Frameworks	01	CO3	PPT	TB1
24.	Models, and Standards.	01	CO3	PPT	TB1
MODULE-4: TEST TOOL STRATEGY(9 Hours)					
25.	Tools to Elicit Examples and Requirements	01	CO4	PPT Open Discussion Session	TB1

S. No	Topic	Contact Hours	CO Mapping	Pedagogy	Resources
26.	Checklists - Mind Maps	01	CO4	Blended Learning	TB1
27.	Spreadsheets - Mock-Ups	01	CO4	PPT and Comparative Study	TB1
28.	Flow Diagrams - Software-Based Tools	01	CO4	PPT and Open Discussion Session	TB1
29.	Tools for Automating Tests Based on Examples	01	CO4	Blended Learning	TB1
30.	Tools to Test below the GUI and API Level	01	CO4	PPT and Case Study	TB1
31.	Tools for Testing through the GUI	01	CO4	PPT and Open Discussion	TB1
32.	Strategies for Writing Tests	01	CO4	PPT and Case Study	TB1
33.	Testability - Code Design and Test Design.	01	CO4	PPT and Open Discussion Session	TB1
MODULE-5: TESTS THAT CRITIQUE THE PRODUCT(9 Hours)					
34.	Introduction to Quadrant 3	01	CO5	PPT and Open Discussion	TB1
35.	Demonstrations - Scenario Testing	01	CO5	PPT and Open Discussion	TB1
36.	Exploratory Testing	01	CO5	PPT and Case Study	TB1
37.	Usability Testing	01	CO5	PPT and Case Study	TB1
38.	Behind the GUI	01	CO5	PPT and Case Study	TB1
39.	Testing Documents and Documentation	01	CO5	PPT and Open Discussion	TB1
40.	User Documentation	01	CO5	PPT and Open Discussion	TB1
41.	Reports	01	CO5	PPT and Open Discussion	TB1
42.	Tools to Assist with Exploratory Testing.	01	CO5	PPT	TB1

COURSE EVALUATION:

S. No.	Evaluation Type	Contents	Duration in Minutes	Marks for Evaluation	Scaling Factor	Max. Marks
1.	Mid Term Exam -1	Module - I & II	90	50	30	30*
2.	Mid Term Exam -2	Module – III, IV & V	90	50	30	
3.	End Term Exam	Full Syllabus	180	100	50	50
4.	Experiential Learning	Content specified above	-	20	-NA-	20
Total Marks						100

* For a total of 30 marks, 80% of better one of the two CIAT and 20% of the other one are added and finalized, any fraction shall be rounded off to the higher integer number.

Code No.: 22CS102010

MODEL QUESTION PAPER



MOHAN BABU UNIVERSITY

Sree Sainath Nagar, Tirupati 517 102

B.Tech. VII Semester (MBU22) Regular Examinations, NOV, 2026
(22CS102010) SOFTWARE TESTING AND QUALITY ASSURANCE
(CSE)

Time: 3 Hours

Max. Marks: 100

PART - A

Answer All Questions.

All Questions Carry Equal Marks

10 x 2 = 20 Marks

1.	a)	Define Agile Testing.	2	BL1 CO1
	b)	List any two principles of Agile Testing.	2	BL1 CO1
	c)	Define Agile Project Team.	2	BL1 CO2
	d)	What is an Independent QA Team?	2	BL1 CO2
	e)	Define Lightweight Process.	2	BL1 CO3
	f)	Define Software Metric.	2	BL1 CO3
	g)	Define Mind Map.	2	BL1 CO4
	h)	What is a Mock-up?	2	BL1 CO4
	i)	Define Exploratory Testing.	2	BL1 CO5
	j)	What is Scenario Testing?	2	BL1 CO5

PART - B

Answer One Question from each Module.

All Questions Carry Equal Marks

5 x 16 = 80 Marks

MODULE-I

2.	a)	Explain Agile Testing and discuss its key characteristics with suitable examples.	7	BL2 CO1
	b)	Compare Traditional Testing and Agile Testing with a neat comparison table.	8	BL3 CO1

(OR)

3.	a)	Explain the principles of Agile Testing and their significance in software quality.	7	BL2	CO1
	b)	Explain the Agile Tester mindset and discuss how Agile values are applied in software testing.	8	BL3	CO1
MODULE-II					
4.	a)	Explain the structure of an Agile Project Team with a neat diagram.	7	BL2	CO2
	b)	Explain the responsibilities of testers in Agile software development.	8	BL2	CO2
(OR)					
5.	a)	Apply Agile team logistics to organize testing activities for an online shopping system.	7	BL3	CO2
	b)	Explain the importance of self-organizing teams in Agile projects.	8	BL3	CO2
MODULE-III					
6.	a)	Explain Lightweight Processes with suitable examples.	8	BL2	CO3
	b)	Explain Software Metrics and classify them with examples.	7	BL2	CO3
(OR)					
7.	a)	Apply Defect Tracking concepts to explain the Defect Life Cycle with a neat diagram.	7	BL3	CO3
	b)	Explain the components of a Test Plan with suitable examples.	8	BL3	CO3
MODULE-IV					
8.	a)	Explain various tools used to elicit examples and requirements.	8	BL2	CO4
	b)	Explain the role of Mind Maps and Mock-ups in Agile Testing.	7	BL2	CO4
(OR)					
9.	a)	Apply suitable testing tools for GUI and API testing in an e-commerce application.	8	BL3	CO4
	b)	Explain software-based testing tools with suitable examples.	7	BL3	CO4
MODULE-V					
10.	a)	Explain Quadrant 3 Testing with suitable examples.	8	BL2	CO5
	b)	Explain Demonstration Testing and Scenario Testing.	7	BL2	CO5
(OR)					

11.	a)	Apply Exploratory Testing techniques to test an online banking application.	8	BL3	CO5
	b)	Explain Usability Testing with suitable examples.	7	BL3	CO5

Target for Course Outcome Attainment:

Course Outcomes	Attainment Target (%)
CO1	60%
CO2	60%
CO3	60%
CO4	60%
CO5	60%

Delivery Details of Content Beyond Syllabus: *(if any)*:

Delivery Details of Content Beyond Syllabus for the Agile Testing course. These topics enhance students' practical knowledge and industry readiness.

Delivery Details of Content Beyond Syllabus

- Introduction to DevOps and Continuous Testing.
- Agile Testing in CI/CD Pipelines using Jenkins and GitHub Actions.
- Test Automation Frameworks using Selenium, Playwright, and Cypress.
- API Testing using Postman and REST Assured.
- Behavior-Driven Development (BDD) using Cucumber.
- Agile Project Management using Jira.
- Version Control and Collaboration using Git and GitHub.
- Introduction to Performance Testing using Apache JMeter.
- Mobile Application Testing using Appium.
- Security Testing Basics and Vulnerability Assessment in Agile Projects.
- AI-assisted Test Case Generation and Test Automation using Generative AI tools.
- Cloud-based Testing Platforms such as BrowserStack and LambdaTest.
- Agile Testing Case Studies from Software Industry Projects.
- Hands-on Mini Project: Designing and Executing an Agile Testing Framework.
- Industry Expert Guest Lecture on Agile QA Practices and Career Opportunities.

These topics provide students with exposure to modern Agile testing practices, automation tools, CI/CD integration, cloud testing, AI-assisted testing, and real-world software quality assurance workflows, making them better prepared for industry roles.

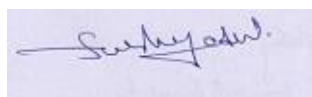
Brief Discussion about other automation testing tools.

Several other testing tools are available to support different types of software testing.

- TestComplete – A commercial tool that automates desktop, web, and mobile application testing using script-based and scriptless approaches.
- Katalon Studio – An easy-to-use automation platform that supports web, mobile, desktop, and API testing with built-in reporting features.
- Robot Framework – An open-source keyword-driven framework used for acceptance testing and robotic process automation (RPA).
- Ranorex Studio – A commercial automation tool for desktop, web, and mobile applications, offering record-and-playback functionality.
- Tricentis Tosca – A model-based automation tool that enables scriptless test automation for enterprise applications.
- UFT One – A commercial functional testing tool that supports GUI, API, and regression testing across multiple platforms.
- SoapUI – A popular tool for functional, regression, and load testing of REST and SOAP web services.
- Apache JMeter – Primarily used for load, stress, and performance testing of web applications and APIs.
- Appium – An open-source framework for automating Android, iOS, and hybrid mobile applications.
- Cypress – A modern JavaScript-based end-to-end testing tool known for fast execution and real-time debugging.
- Playwright – Supports cross-browser automation with reliable testing for modern web applications.
- WebdriverIO – A Node.js-based framework for web and mobile application testing with support for multiple automation services.

SDG Mapping: *(Identify the Sustainable Development Goals addressed by the course).*

SDG No	Relevance
SDG 4: Quality Education	Develops students' knowledge and practical skills in Agile software testing, quality assurance, test automation, and collaborative software development, thereby improving employability and lifelong learning.
SDG 8: Decent Work and Economic Growth	Prepares graduates with industry-relevant testing and quality assurance skills that improve software quality, productivity, innovation, and workforce readiness.
SDG 9: Industry, Innovation and Infrastructure	Promotes the use of Agile methodologies, modern testing tools, automation frameworks, and software quality practices that support reliable and innovative digital infrastructure.
SDG 12: Responsible Consumption and Production)	Encourages efficient software development through defect prevention, continuous testing, quality improvement, and optimal utilization of testing resources, reducing software rework and waste.
SDG 17: Partnerships for the Goals	Emphasizes teamwork, collaboration among developers, testers, stakeholders, and cross-functional Agile teams to achieve high-quality software products.



Signature of the Course Instructor



Signature of the Chairperson BOS